

EE204 Semiconductor Devices

HW 3

1.

A pure silicon sample maintained at room temperature has an intrinsic carrier concentration of $1.5 \times 10^{10} \text{ cm}^{-3}$. It is first doped with donors of concentration $2 \times 10^{14} \text{ cm}^{-3}$, followed by acceptors of concentration $4 \times 10^{14} \text{ cm}^{-3}$. Assuming that the carrier mobilities are $\mu_n = 1350 \text{ cm}^2/\text{Vs}$ and $\mu_p = 480 \text{ cm}^2/\text{Vs}$,

- calculate the majority and minority carrier concentrations,
- what is the resistivity of the pure sample, prior to the two types of dopings?
- how will the resistivity change after the dopings?

$$[2 \times 10^{14} \text{ cm}^{-3}, 1.125 \times 10^6 \text{ cm}^{-3}, 2 \times 10^5 \Omega\text{-cm}, 65 \Omega\text{-cm}]$$

2.

The electron concentration in silicon at $T = 300 \text{ K}$ is given by

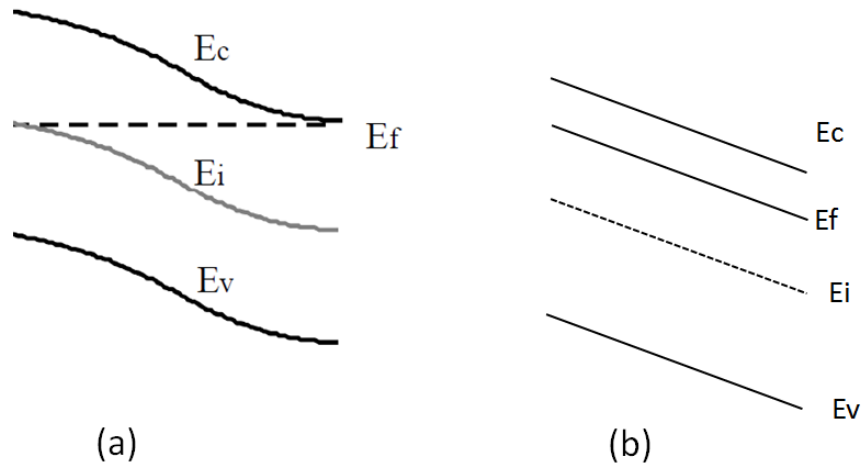
$$n(x) = 10^{16} \exp\left(\frac{-x}{18}\right) \text{ cm}^{-3}$$

where x is measured in μm and is limited to $0 \leq x \leq 25 \mu\text{m}$. The electron diffusion coefficient is $D_n = 25 \text{ cm}^2/\text{s}$ and the electron mobility is $\mu_n = 960 \text{ cm}^2/\text{V-s}$. The total electron current density through the semiconductor is constant and equal to $J_n = -40 \text{ A/cm}^2$. The electron current has both diffusion and drift current components. Determine the electric field as a function of x which must exist in the semiconductor.

$$E = 14.5 - (26.0) \exp\left(\frac{+x}{18}\right)$$

3.

For below band diagrams, what are the differences between them?
Specify the differences in terms of doping, electrical field, equilibrium etc.



4.

A p-type silicon sample has an acceptor doping concentration of 10^{16} cm^{-3} . At $t < 0$, it is uniformly irradiated with light of an appropriate wavelength resulting in the generation of electron-hole pairs at a rate of $G_L = 10^{17} \text{ cm}^{-3}\text{s}^{-1}$. For a temperature of 300 K,

- assuming a minority carrier lifetime of $\tau_0 = 10 \text{ } \mu\text{s}$, calculate the steady state excess carrier concentrations,
- determine the quasi-Fermi energy levels and indicate their positions in an energy band diagram.

Take $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$.

$$[10^{12} \text{ cm}^{-3}, 0.347 \text{ eV}, 0.109 \text{ eV}]$$

5.

Consider silicon at $T = 300 \text{ K}$ that is doped with donor impurity atoms to a concentration of $N_d = 5 \times 10^{15} \text{ cm}^{-3}$. The excess carrier lifetime is $2 \times 10^{-7} \text{ s}$. (a) Determine the thermal equilibrium recombination rate of holes. (b) Excess carriers are generated such that $\delta n = \delta p = 10^{14} \text{ cm}^{-3}$. What is the recombination rate of holes for this condition?

$$[2.25 \times 10^{11} \text{ cm}^{-3}\text{s}^{-1}, 5 \times 10^{20} \text{ cm}^{-3}\text{s}^{-1}]$$

6.

An n-type silicon sample contains a donor concentration of $N_d = 10^{16} \text{ cm}^{-3}$. The minority carrier hole lifetime is found to be $\tau_{p0} = 20 \mu\text{s}$. (a) What is the lifetime of the majority carrier electrons? (b) Determine the thermal-equilibrium generation rate for electrons and holes in this material. (c) Determine the thermal-equilibrium recombination rate for electrons and holes in this material.

$$[8.89 \times 10^6 \text{ s}, 1.125 \times 10^9 \text{ cm}^{-3}\text{s}^{-1}]$$

7.

A piece of p-type doped silicon at 300 K has an acceptor concentration of $5 \times 10^{14} \text{ cm}^{-3}$. Assume excess carriers are present and that $E_F - E_{Fpq} = 0.01 k_B T$ and intrinsic concentration of $1.5 \times 10^{10} \text{ cm}^{-3}$.

(a) Does this condition correspond to low injection?

(b) Determine $E_{Fnq} - E_i$.

$$[\text{Yes}, 0.150 \text{ eV}]$$

8.

A silicon sample at 300 K is n-type with $N_d = 5 \times 10^{16} \text{ cm}^{-3}$ and $N_a = 0$. The sample has a length of 0.1 cm and a cross-sectional area of 10^{-4} cm^2 . A voltage of 5V is applied between the ends of the sample. For $t < 0$, the sample has been illuminated with light, producing an excess-carrier generation rate of $5 \times 10^{21} \text{ cm}^{-3}\text{s}^{-1}$ uniformly throughout the entire silicon. The minority carrier lifetime is $0.3 \mu\text{s}$. At $t = 0$, the light is turned off. Derive the expression for the current in the sample as a function of time $t \geq 0$. Take $\mu_n = 1350 \text{ cm}^2/\text{Vs}$, $\mu_p = 480 \text{ cm}^2/\text{Vs}$ and $n_i = 1.5 \times 10^{10}/\text{cm}^3$

$$[54 + 2.2 \exp(-t/\tau_{p0}) \text{ mA}]$$

9.

Consider a semi-infinite p-type silicon bar doped homogeneously to a value of $1.39 \times 10^{16}/\text{cm}^3$ at 300 K. The intrinsic carrier concentration is $1.5 \times 10^{10}/\text{cm}^3$. The applied electric field is zero. A minority carrier concentration is electrically injected at one end of the sample ($x = 0$) such that the excess minority carrier concentration at $x = 0$ is $2.49 \times 10^{13}/\text{cm}^3$. The electron mobility and the electron lifetime are $1450 \text{ cm}^2/\text{V-s}$ and 10^{-6} s respectively.

i) Calculate the steady state excess electron concentration as function of x

ii) Calculate the electron diffusion current density at $x = 0$ and $x = 10 \mu\text{m}$

$$[2.49 \times 10^{13} \exp(-x/61.4 \times 10^{-4})/\text{cm}^3, 24.5 \text{ mA}/\text{cm}^2, 20.8 \text{ mA}/\text{cm}^2]$$

10.

Consider a semi-infinite n-type silicon bar doped homogeneously to a value of $2.16 \times 10^{16}/\text{cm}^3$ at 300 K. The intrinsic carrier concentration is $1.5 \times 10^{10}/\text{cm}^3$. The applied electric field is zero. A minority carrier concentration is electrically injected at one end of the sample ($x = 0$) such that the excess minority carrier concentration at $x = 0$ is $1.6 \times 10^{13}/\text{cm}^3$. The hole mobility and the hole lifetime are $450 \text{ cm}^2/\text{V-s}$ and 10^{-8} s respectively.

i) Calculate the steady state excess hole concentration as function of x

ii) Calculate the hole diffusion current density at $x = 0$.

$$[1.6 \times 10^{13} \exp(-x/3.42 \times 10^{-4})/\text{cm}^3, 87.6 \text{ mA}/\text{cm}^2]$$